Rates of Change and Rectilinear Motion

Section 1: Position, Velocity & Speed

1 A particle moves along the x-axis with position $x(t) = t^3 - 6t^2 + 9t$ for $t \geq 0$. At what times is the particle at rest?

a. $t = 1$ and $t = 3$

b. $t = 0$ and $t = 3$

c. $t = 2$

d. $t = 3$ only

2 What is the speed of the particle when $v(t) = -8$ and $a(t) = 3$?

a. 8

b. -8

c. 5

d. $\sqrt{73}$

3 When is a particle's speed increasing?

a. When velocity and acceleration have the same sign

b. When velocity is positive

c. When acceleration is positive

d. When acceleration is zero

Section 2: Total Distance vs. Displacement

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4 If a particle's position is $s(t) = t^2 - 4t$ on $[0, 3]$, what is its net displacement?

a. -3

b. 5

c. -4

d. 3

5 For the above particle with $s(t) = t^2 - 4t$ on $[0, 3]$, what is the total distance traveled?

a. 5

b. 3

c. 4

d. 7

6 Which integral represents the total distance traveled by a particle from $t = a$ to $t = b$?

a. $\int_a^b |v(t)|dt$

b. $\int_a^b v(t)dt$

c. $|\int_a^b v(t)dt|$

d. $\int_a^b a(t)dt$

Section 3: Related Rates: Geometric Models

7 The radius of a circle increases at a constant rate of 3 cm/s. How fast is the area increasing when the radius is 10 cm?

a. 60π cm^2/s

b. 30π cm^2/s

c. 100π cm^2/s

d. 20π cm^2/s

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8 A 13-foot ladder leans against a vertical wall. If the bottom slides away from the wall at 2 ft/s, how fast is the top sliding down when the bottom is 5 feet from the wall?

a. $-\frac{5}{6}$ ft/s

b. $-\frac{12}{5}$ ft/s

c. -2 ft/s

d. $-\frac{5}{12}$ ft/s

9 Air is pumped into a spherical balloon at a rate of 100 cm^3/s. How fast is the radius increasing when $r = 5$ cm?

a. $\frac{1}{\pi}$ cm/s

b. $\frac{1}{4\pi}$ cm/s

c. $\frac{4}{\pi}$ cm/s

d. 25π cm/s

Section 4: Related Rates: Angles & Proportions

10 A conical tank with vertex down has height 10 m and base radius 4 m. Water flows in at 2 m^3/min. What is the relation between r and h?

a. $r = \frac{2}{5}h$

b. $r = \frac{5}{2}h$

c. $r = \frac{1}{2}h$

d. $r = 4h$

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11 Using $V = \frac{1}{3}\pi r^2 h = \frac{4}{75}\pi h^3$, how fast is the water level rising when $h = 5$ m and $\frac{dV}{dt} = 2$ m^3/min?

a. $\frac{1}{2\pi}$ m/min

b. $\frac{2}{\pi}$ m/min

c. $\frac{1}{4\pi}$ m/min

d. $\frac{4}{25\pi}$ m/min

12 A street light is mounted at the top of a 15-foot pole. A 6-foot man walks away from the pole at 5 ft/s. How fast is the tip of his shadow moving?

a. $\frac{25}{3}$ ft/s

b. 5 ft/s

c. $\frac{10}{3}$ ft/s

d. 10 ft/s

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